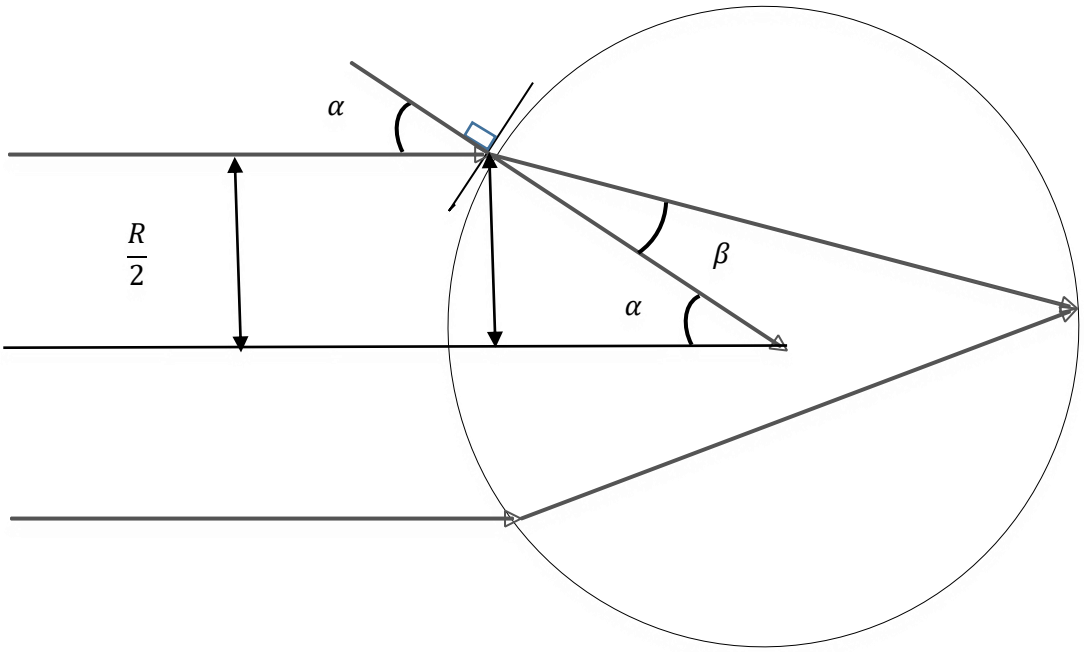
1-Masala



Rasmdan ko'rinib turibdiki α burchak 30 ° qarshisida yotgan katet gipotenuzaning yarmiga teng aylana qoidasiga ko'ra esa

$\beta = \frac{\alpha}{2}$ demak β burchak 15°ga teng ekan.

Sinish qonuniga ko'ra

$n = \frac{\sin\alpha}{\sin\beta}$ ga teng

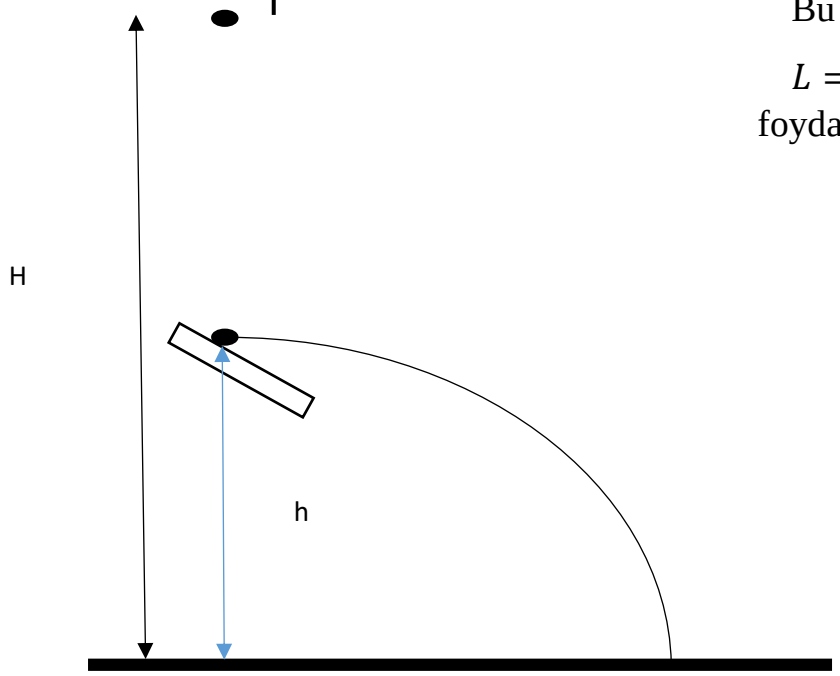
$n = \frac{\sin 30°}{\sin 15°} = 2cos15° = 1.93$ javob: $n = 1{,}93$

2- MASALA

Berilgan
H=25m
h=16m
$\alpha = 45°$
L=?

Erkin tushish qoidasiga ko'ra jism qiyalikka urilguncha $v = \sqrt{2g(H-h)}$ tezlikka erishadi

yo'lning qolgan qismida ham jism y o'qi bo'ylab erkin Tushadi va yerga $t = \sqrt{\frac{2h}{g}}$ vaqtda tushadi



Bu ikki tenglamadan

$L = v * t$ ekanligidan foydalansak

$$L = 2\sqrt{h * (H-h)}$$

Ifoda kelib chiqadi

Berilganlarni o'rniga qo'yib hisoblasak

$$L = 2\sqrt{16 * (25-16)}\ [m]$$

$L = 24\ [m]$ ekanligi kelib chiqadi

Javob: $L = 24\ [m]$

3- Masala

Berilgan

$v_1 = 1\ m/s$

$v_2 = 2\ m/s$

$v_3 = 3\ m/s$

v_n =?

Qarshilik kuchi tezlikka proporsional.

$F_q = k * v$ vaqt o'tishi bilan tezlik o'zgaradi va o'rtacha quvvat $N = \frac{k\,v^2}{2}$ bo'ladi

Har bitta motor uchun bu ifodani yoziv chiqsak

$$N_1 = \frac{k*{v_1}^2}{2}$$

$$N_2 = \frac{k*{v_2}^2}{2}$$

$$N_3 = \frac{k*{v_3}^2}{2}$$

Agar uchala motor katerga qo'yilsa umumiy quvvat har bir quvvatning yig'indisi bo'ladi.

$$N_{um} = N_1 + N_2 + N_3 = k * \frac{{v_n}^2}{2}$$

$$N_{um} = \frac{k * {v_1}^2}{2} + \frac{k * {v_2}^2}{2} + \frac{k * {v_3}^2}{2} = k * \frac{{v_n}^2}{2}$$

$$v_n = \sqrt{{v_1}^2 + {v_2}^2 + {v_3}^2}$$

$$v_n = \sqrt{14} \quad \left[\frac{m}{s}\right]$$

Javob : $v_n = \sqrt{14} \quad \left[\frac{m}{s}\right]$